

Nainika Roy

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EDUCATION

Columbia University in the City of New York	2018- present
<i>Ph.D. Student in Neurobiology & Behavior</i>	
<i>M.Phil in Neurobiology & Behavior</i>	2018-2021
University of Illinois at Urbana-Champaign (UIUC)	2013-2017
<i>Bachelor of Science in Molecular and Cellular Biology with High Distinction</i>	
<i>Minor in Informatics</i>	

RESEARCH & WORK EXPERIENCE

Columbia University (New York City, NY)	2018- present
<i>Doctoral Candidate (Sher Lab, Center for Translational and Computational Neuroimmunology)</i>	
Project: Identifying novel drivers of microglial dysfunction in Alzheimer's disease	
• Investigating the function of Alzheimer's disease-associated immune genes through a multipronged approach utilizing precision genomic editing of human cell lines, histopathological analysis of patient brain samples and other assays to robustly understand the role of microglia in disease pathobiology	
European Molecular Biology Laboratory Rome (Monterotondo, Italy)	2017- present
<i>Research Intern (Heppenstall Group)</i>	
Project: Molecule guided laser ablation as a novel therapeutic strategy to control itch and inflammatory pain	
• Utilized mouse models of diseases such as atopic dermatitis, FPLCA, spared nerve injury, and osteoarthritis to evaluate effectiveness of a novel, developed treatment and evaluated results through various behavioral and molecular assays	
Beckman Institute for Advanced Science and Technology, UIUC (Urbana, IL)	2015- 2017
<i>Undergraduate Research Assistant (OpNTech Laboratory)</i>	
Project: Effects of a novel $\alpha 7$ nicotinic acetylcholine receptor agonist on neuronal activity evaluated by multi-electrode electrophysiology	
• Studied the effects of a receptor agonist on neuronal networks through analysis of electrophysiological data	
University of Texas Health Science Center at Houston (Houston, TX)	2016
<i>Summer Undergraduate Research Trainee (De Lay Lab)</i>	
Project: Testing the role of PNPase – RNase E interactions on sRNA- mediated gene regulation in <i>E. coli</i>	
• Identified protein interaction disrupting mutants, examined the effect of reduced interaction on sRNA activity	
National Center for Supercomputing Application (Urbana, IL)	2016- 2017
<i>SPIN (Students Pushing Innovation) Intern (High Performance Computing Biology group)</i>	
Project: Assessing, documenting and compiling current bioinformatics data formats used in genome analysis workflow	
LAS 101 Teaching Assistant, UIUC (Urbana, IL)	2016- 2017
• Designed, implemented and taught a weekly class of 25 students about academic success, served as a mentor	
• Attended curriculum discussions, leadership development seminars and workshops	
Social Sciences, Health and Education Library Research Assistant, UIUC (Urbana, IL)	2015- 2016
Student Coordinator, Illinois Summer Neuroscience Institute (ISNI), UIUC (Urbana, IL)	2015

OTHER POSITIONS & COMMUNITY ENGAGEMENT

Mentor, Polygence	2020- present
Founder & Organizer, Columbia Access Neuroscience	2020- 2022

Writer, Scientists on the Subway	2020- 2022
Scientific Editor, Cactus Communications	2019- present
Volunteer, World Science Festival	2019- present
Volunteer Teacher, Single-visit Classroom Teaching Program, Columbia University	2019- present
Volunteer, REACT Outreach Program	2015- 2016

RECOGNITION

AY21 DEI Service Award 2021

- In recognition of exceptional service towards Diversity, Equity, and Inclusion initiatives in department

PUBLICATIONS

1. Linda Nocchi, Nainika Roy, Mariangela D'Attilia, Rahul Dhandapani, Mariano Maffei, Andrei Traista, Laura Castaldi, Emerald Perlas Cora Hallie Chadick & Paul A. Heppenstall. Interleukin-31-mediated photoablation of pruritogenic epidermal neurons reduces itch-associated behaviors in mice. *Nature Biomedical Engineering* (2018)
2. Linda Nocchi, Carla Portulano, Federica Franciosa, Balint Doleschall, Mirela Panea, Nainika Roy, Mariano Maffei, Alessandra Gargano, Emerald Perlas, & Paul Heppenstall. NGF-mediated photoablation of nociceptors reduces pain behavior in mice. *Pain* (2019)